

Class 9: Candy Mini-Project

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Background

Today we are taking a detour to analyze a fun dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA)

```
candy_file <- "candy-data.csv"

candy = read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Data import

The data is all related to Halloween candy and is from the 538 website.

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
library(tidyverse)
```

```
candy %>%  
  filter(fruity == T) %>%  
  nrow()
```

```
[1] 38
```

Favorite candy

One of the most interesting variables in the dataset is winpercent. For a given candy this value is the percentage of people who prefer this candy over another randomly chosen candy from the dataset (what 538 term a matchup). Higher values indicate a more popular candy.

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
candy |>  
  filter(row.names(candy)=="Twix") |>  
  select(winpercent)
```

```
      winpercent  
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy |>  
  filter(row.names(candy)=="Skittles original") |>  
  select(winpercent)
```

```
      winpercent
Skittles original  63.08514
```

Q4. What is the winpercent value for “Kit Kat”?

```
candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
      winpercent
Kit Kat      76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy |>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```
      winpercent
Tootsie Roll Snack Bars  49.6535
```

skimr::skim() function

There is a useful `skim()` function in the **skimr** package that can help give you a quick overview of a given dataset. Let’s install this package and try it on our candy data.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The winpercent mean is on a different scale. Every other variable is below 1, where win percent is at 50.32%

Q7. What do you think a zero and one represent for the `candy$chocolate` column?

A zero and one represents whether or not a candy contains chocolate or not.

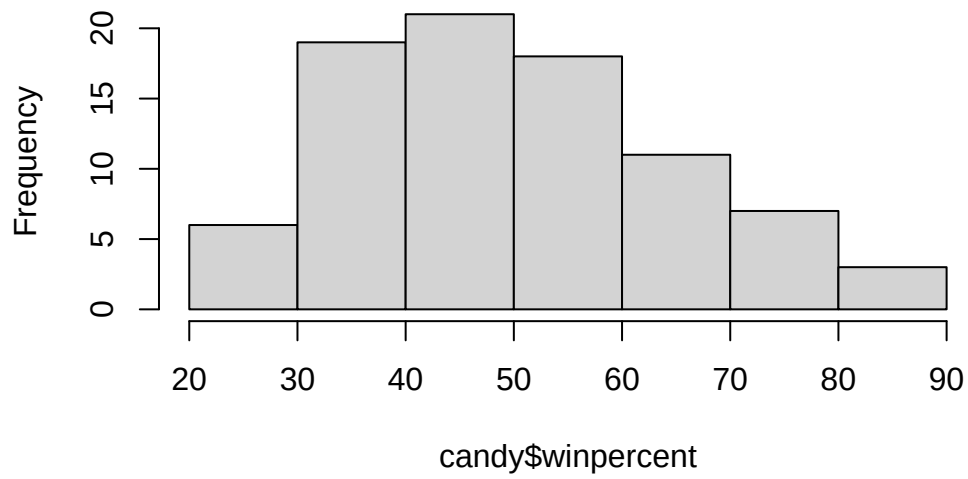
Exploratory Analysis

A good place to start any exploratory analysis is with a histogram. You can do this most easily with the base R function `hist()`. Alternatively, you can use `ggplot()` with `geom_hist()`. Either works well in this case and (as always) its your choice.

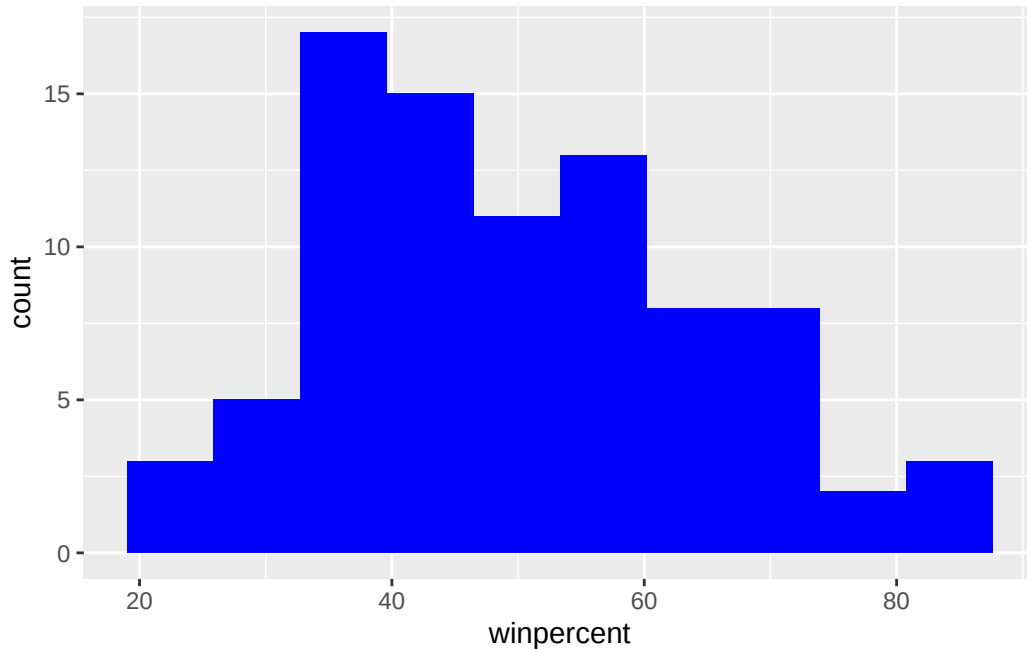
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
## Base R
hist(candy$winpercent)
```

Histogram of candy\$winpercent



```
##ggplot2
ggplot(candy,
  aes(winpercent)) +
  geom_histogram(bins=10, fill="blue")
```



Q9. Is the distribution of winpercent values symmetrical?

The distribution of winpercent values is not symmetrical.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

The center of the distribution is above 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
## chocolate candy win %
chocolatewin <- candy$winpercent[as.logical(candy$chocolate)]
## fruit candy win %
fruitywin <- candy$winpercent[as.logical(candy$fruity)]

## testing whether chocolate average win % > fruit candy win %
mean(chocolatewin) > mean(fruitywin)
```

```
[1] TRUE
```

On average, chocolate candy is ranked higher than fruit candy.

Q12. Is this difference statistically significant?

```
t.test(chocolatewin, fruitywin)
```

Welch Two Sample t-test

```
data: chocolatewin and fruitywin
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Given that the p-value is significantly below the 95% confidence interval ($\alpha=0.05$), the null hypothesis is rejected, so there is a statistical significance between the win percentage between chocolate and fruit candy.

Overall Candy Rankings

Let's use the base R `order()` function together with `head()` to sort the whole dataset by a variable/column of interest. Or if you have been getting into the tidyverse and the `dplyr` package you can use the `arrange()` function together with `head()` to do the same thing and answer the following questions:

Q13. What are the five least liked candy types in this set?

```
candy %>%
  arrange(winpercent) %>%
  head(5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard	bar	pluribus	sugarpercent	pricepercent
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

```
candy %>%
  arrange(winpercent) %>%
  tail(5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Snickers	1	0	1	1	1
Kit Kat	1	0	0	0	0
Twix	1	0	1	0	0
Reese's Miniatures	1	0	0	1	0
Reese's Peanut Butter cup	1	0	0	1	0

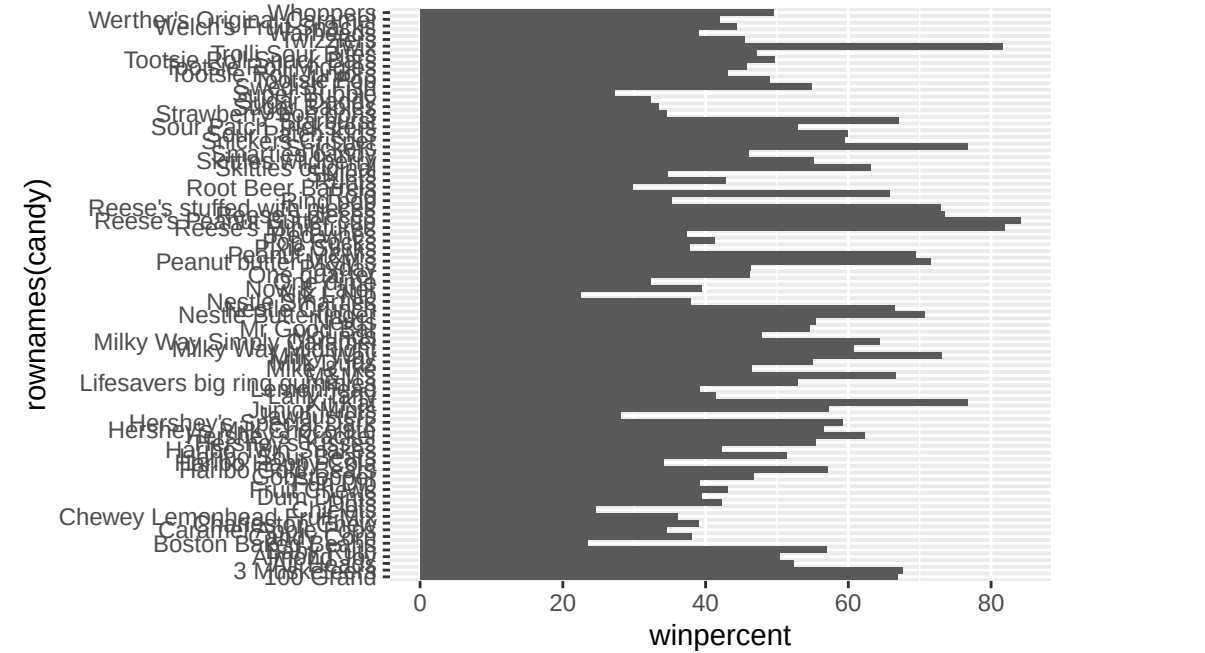
	crispedricewafer	hard	bar	pluribus	sugarpercent
Snickers	0	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Twix	1	0	1	0	0.546
Reese's Miniatures	0	0	0	0	0.034
Reese's Peanut Butter cup	0	0	0	0	0.720

	pricepercent	winpercent
Snickers	0.651	76.67378
Kit Kat	0.511	76.76860
Twix	0.906	81.64291
Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029

To examine more of the dataset in this vain we can make a barplot to visualize the overall rankings. As always we will use an iterative approach to building a useful visualization by getting a rough starting plot and then refining and adding useful details in a stepwise process.

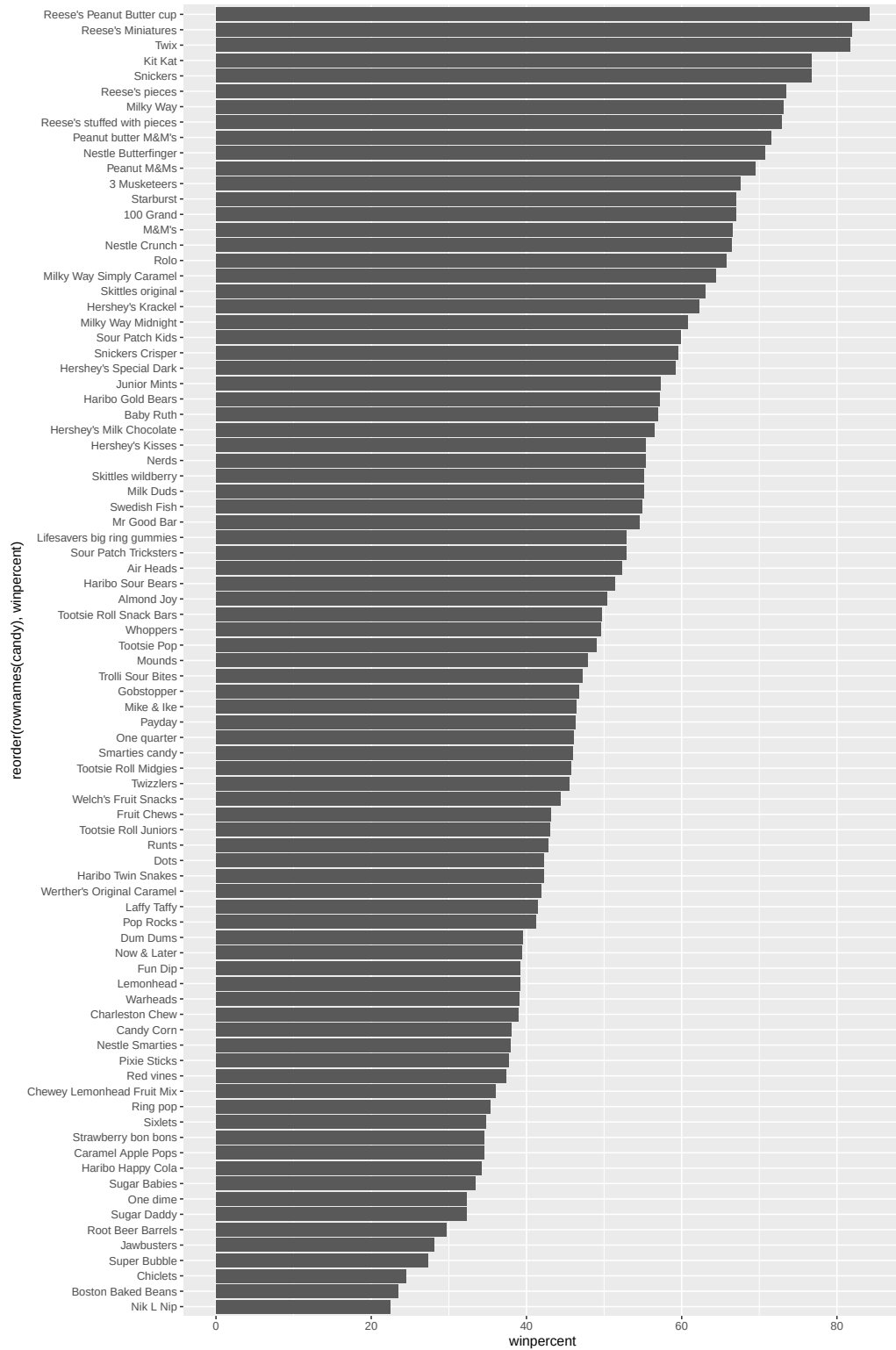
Q15. Make a first barplot of candy ranking based on winpercent values.


```
ggplot(candy,
       (aes(winpercent, rownames(candy)))) +
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy,
       (aes(winpercent,
             reorder(rownames(candy), winpercent)))) +
geom_col()
```



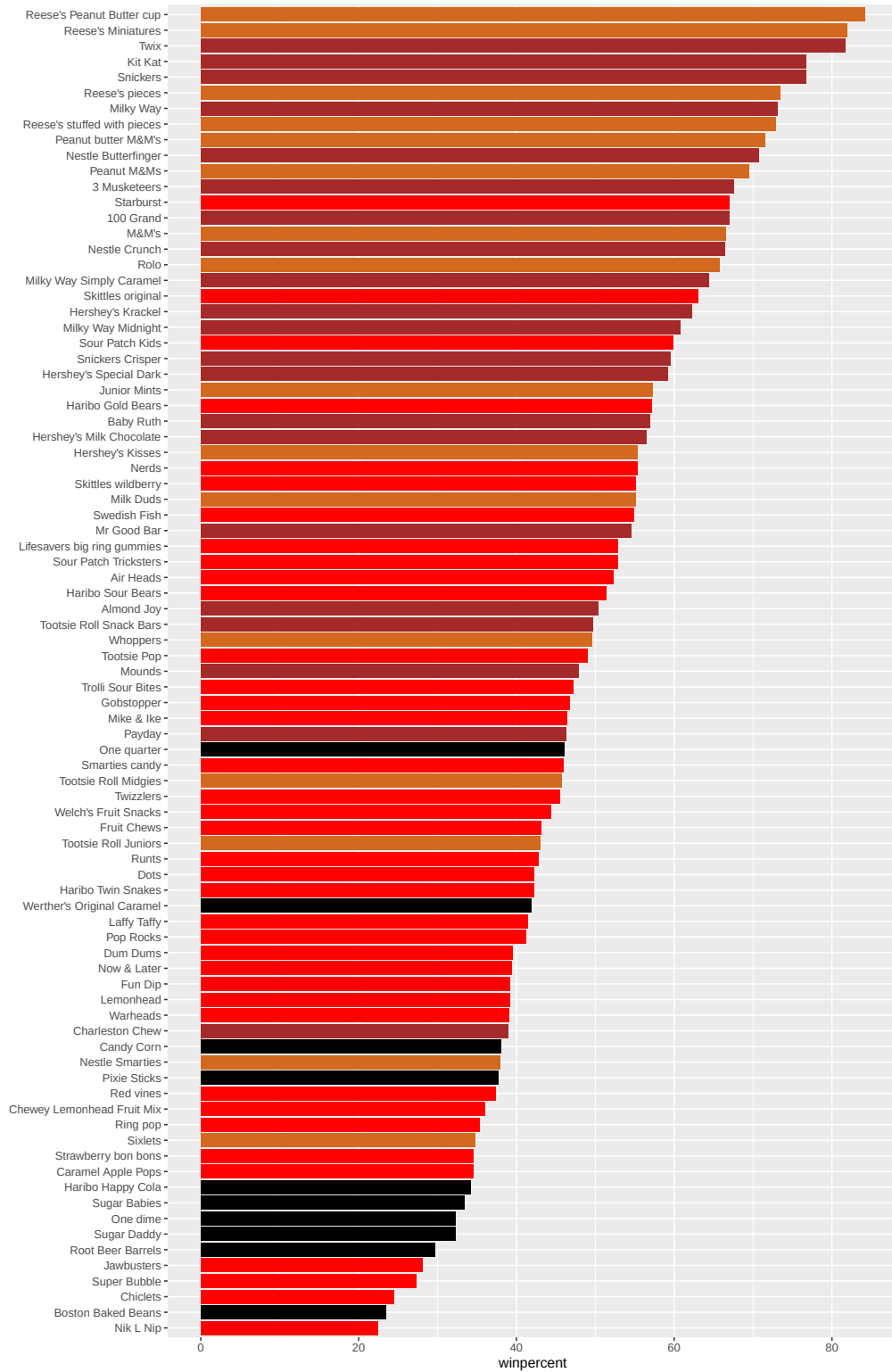
Adding useful color

Let's setup a color vector (that signifies candy type) that we can then use for some future plots. We start by making a vector of all black values (one for each candy). Then we overwrite chocolate (for chocolate candy), brown (for candy bars) and red (for fruity candy) values.

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "red"
```

Now let's try our barplot with these colors. Note that we use `fill=my_cols` for `geom_col()`. Experiment to see what happens if you use `col=mycols`.

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is “Sixlets”.

Q18. What is the best ranked fruity candy?

The best ranked fruity candy is “Starburst”.

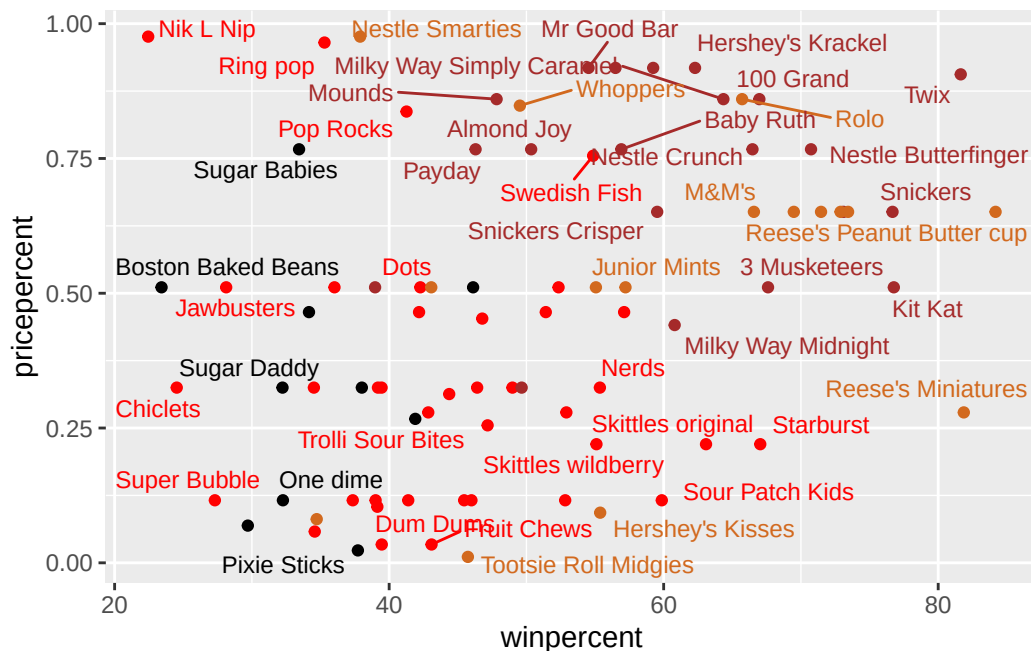
Taking a look at pricepercent

What about value for money? What is the best candy for the least money? One way to get at this would be to make a plot of winpercent vs the pricepercent variable. The pricepercent variable records the percentile rank of the candy’s price against all the other candies in the dataset. Lower values are less expensive and higher values are more expensive.

To this plot we will add text labels so we can more easily identify a given candy. There is a regular `geom_label()` that comes with `ggplot2`. However, as there are quite a few candies in our dataset lots of these labels will be overlapping and hard to read. To help with this we can use the `geom_text_repel()` function from the `ggrepel` package.

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 10)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
candy %>%
  mutate(wintomoney = winpercent/pricepercent) %>%
  arrange(desc(wintomoney)) %>%
  head(1)
```

	chocolate	fruity	caramel	peanuty	almondy	nougat	
Tootsie Roll Midgies	1	0	0	0	0	0	
	crisped	rice	wafer	hard	bar	pluribus	sugarpercent
Tootsie Roll Midgies		0	0	0	1	0.174	
	pricepercent	winpercent	wintomoney				
Tootsie Roll Midgies	0.011	45.73675	4157.886				

The chocolate candy, “Tootsie Roll Midgies”, is the highest ranked in terms of winpercent for the least money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
candy %>%
  arrange(desc(pricepercent)) %>%
  select(pricepercent, winpercent) %>%
  head(5)
```

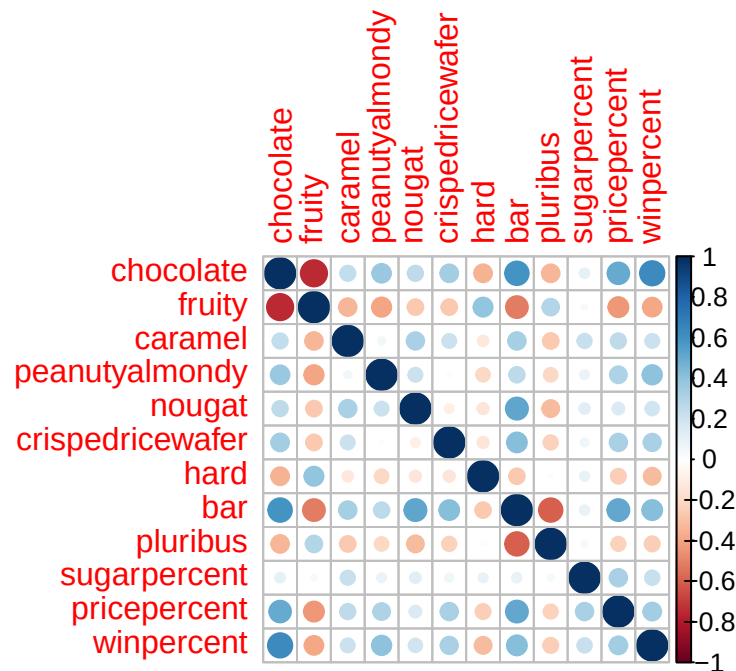
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Exploring the Correlation Structure

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The variables chocolate and fruity are anti-correlated.

Q23. Use your corplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

I predict that chocolate and fruit will have large contributions to PC1.

Principal Component Analysis

```
pca <- prcomp(candy, scale=T)
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

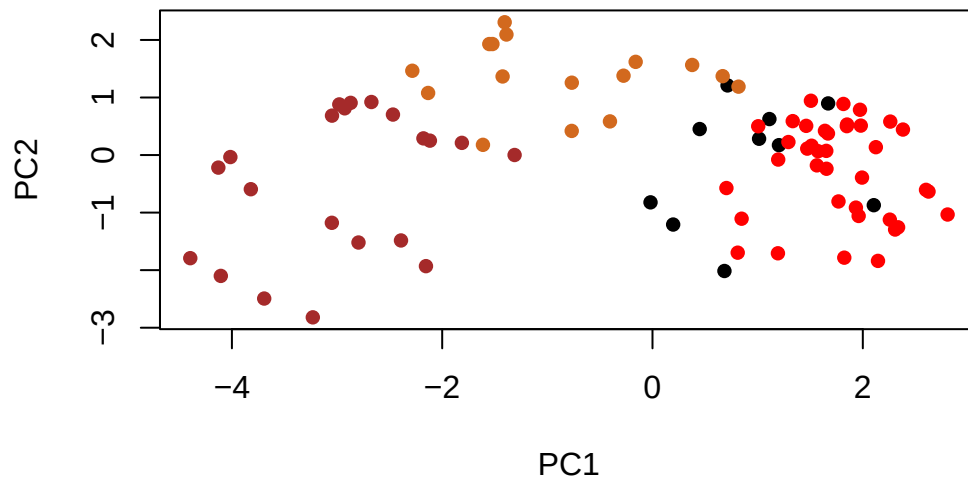
	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
attributes(pca)
```

```
$names
[1] "sdev"      "rotation" "center"    "scale"     "x"
```

```
$class
[1] "prcomp"
```

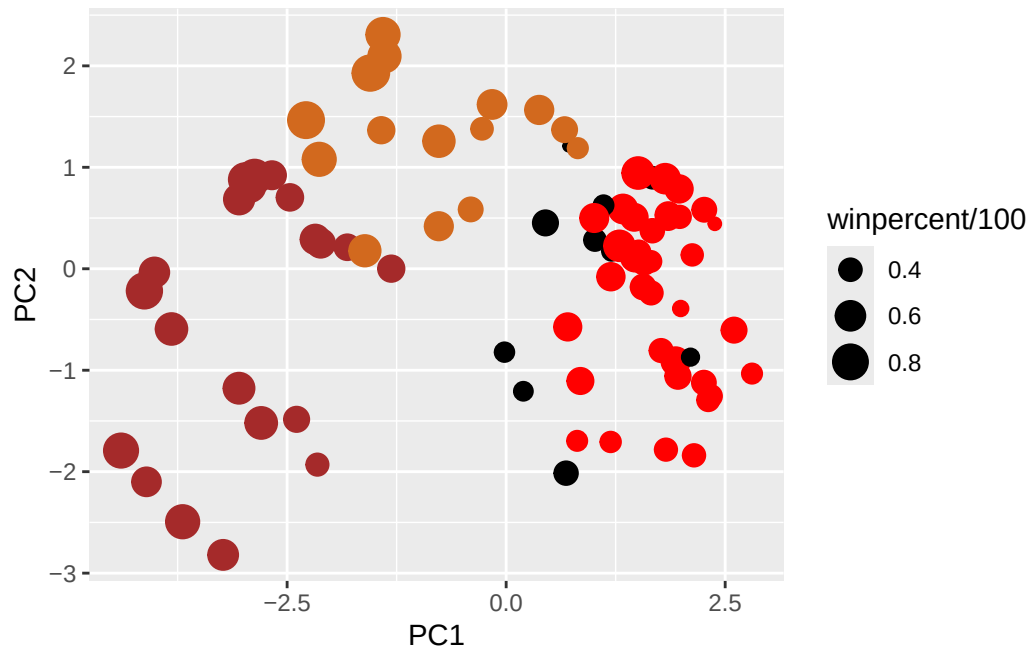
```
plot(pca$x[,1:2], col=my_cols, pch=16)
```

```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])

p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)

p
```

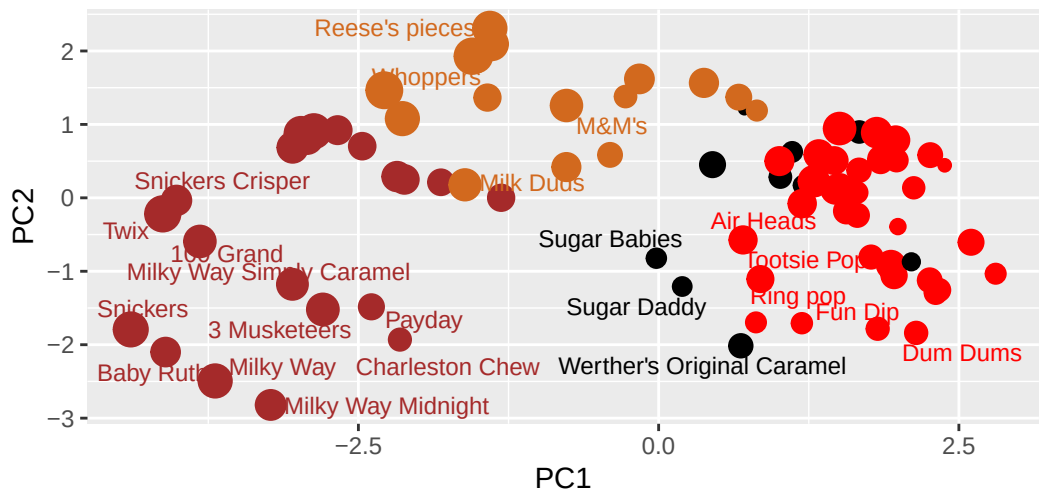


```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate
other (light brown), fruity (red), other (black)",
        caption="Data from 538")
```

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown), fruity (red), other (black)

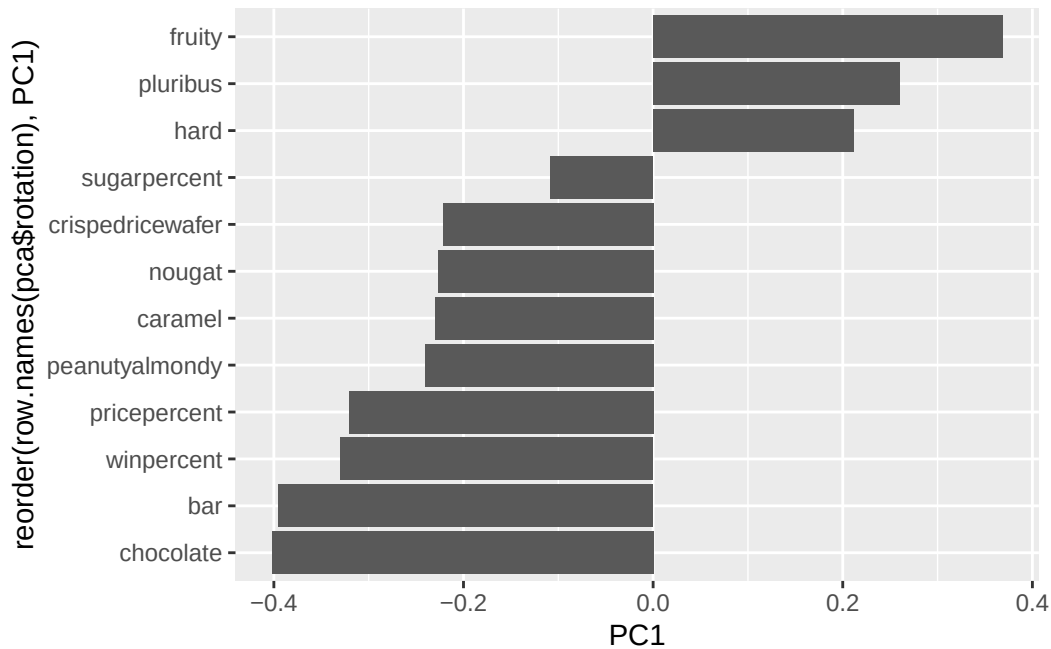


Data from 538

```
##library(plotly)
##ggplotly(p)
```

Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

```
ggplot(pca$rotation) +
  aes(PC1, reorder(row.names(pca$rotation), PC1))+
  geom_col()
```



The variables that are strongly picked up by PC1 in the positive direction are fruity, pluribus, and hard. These make sense, as they have strong positive correlations shown in the earlier correlation matrix plot.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Based on the results found, a winning candy seems to be a bar candy that is made of chocolate. From the data visualization and correlations, chocolate bar candies consistently rank high, and the PCA also demonstrates this, as the two variables of bar and chocolate shows a strong influence while being correlated with win percent.